HACK4HACK CREW HIRING CHALLENGE 2026 · PROBLEM STATEMENT 1

CrewForge

Race-proof team formation at scale

A database-enforced membership workflow designed for the last-seat race.

50-CONTENDER TEST

65 TESTS PASS

100 VU BENCH

Repository: github.com/PSP-YN/Race-proof-team-formation

NOV 2026

The failure mode is a race, not a button

Last seat contention

Many participants can request the same final slot at nearly the same instant.

Duplicate membership

One person must never become a member of two teams in the same hackathon.

Retry ambiguity

A double-click or flaky network must not create duplicate requests or teams.

Correctness invariant

The database—not the browser—decides who owns a seat. Capacity, membership, request status and audit history must commit together.

Target scale: 10,000 participants · 2,000 teams · 100 concurrent board users

A moderated workflow keeps team ownership explicit

01

Discover

Searchable team board with skill-fit recommendations and clear open-seat counts.

→

02

Request

Participant submits an idempotent pending request; no seat is reserved.

→

03

Approve

Only the current leader can accept or decline a request.

→

04

Maintain

Leader can remove a member; leadership passes when the leader leaves.

Invite lifecycle · Expiring, single-use hashed tokens · Final-member exit closes the team · Every mutation is audited

Correctness is enforced at the transaction boundary

Next.js API
Auth · validation · rate limits

→

MongoDB txn
Conditional capacity update

→

Unique indexes
One member / hackathon

→

Audit + SSE
Commit event · live refresh

Atomic approval transaction

Verify leader + pending request → insert membership → increment memberCount only below maxSize → mark request accepted → append audit event

MongoDB replica-set transactions and change streams are required. Unique compound indexes remain the final guard against duplicate membership.

50 contenders. One remaining seat. Exactly one approval.

50
pending join requests

→

50
concurrent leader approvals

→

1
accepted

49
rejected

Assertions from the MongoDB replica-set integration test

2 total memberships (leader + winner) · memberCount = 2 · exactly 1 JOIN_REQUEST_ACCEPTED audit event

Test exercises production requestJoinTransaction and decideJoinRequestTransaction helpers—not a mock race-only helper.

Measured evidence is reproducible—and scoped

The same CI run provisions MongoDB, checks the race invariant, builds the app and exercises the board under load.

65 / 65
tests passed

45
API route-contract tests

11
MongoDB integration tests

9
unit tests

CI LOAD TEST · 100 VIRTUAL USERS · 30 SECONDS

3,000 requests **97.43 req/s** **20.92 ms p95** **0% HTTP failures**

6,000 / 6,000 board checks passed. The same run verified 10,000 seeded participants, 2,000 teams and 10,000 memberships, then ran the 50-contender final-seat approval race.

Environment: GitHub-hosted runner + disposable MongoDB 7 replica set. This is a regression benchmark, not a production-region capacity claim.

CI 41873925487 · CodeQL 41873925485 · Commit a809e7d

Security and lifecycle controls

Identity & authorization

HttpOnly session cookies; server-side leader checks; same-origin mutation checks.

Abuse resistance

Persistent rate limits; high-entropy invite tokens stored as hashes; input validation.

Auditability

Transactional audit events; hashed single-use auth tokens; security telemetry.

Realtime and lifecycle

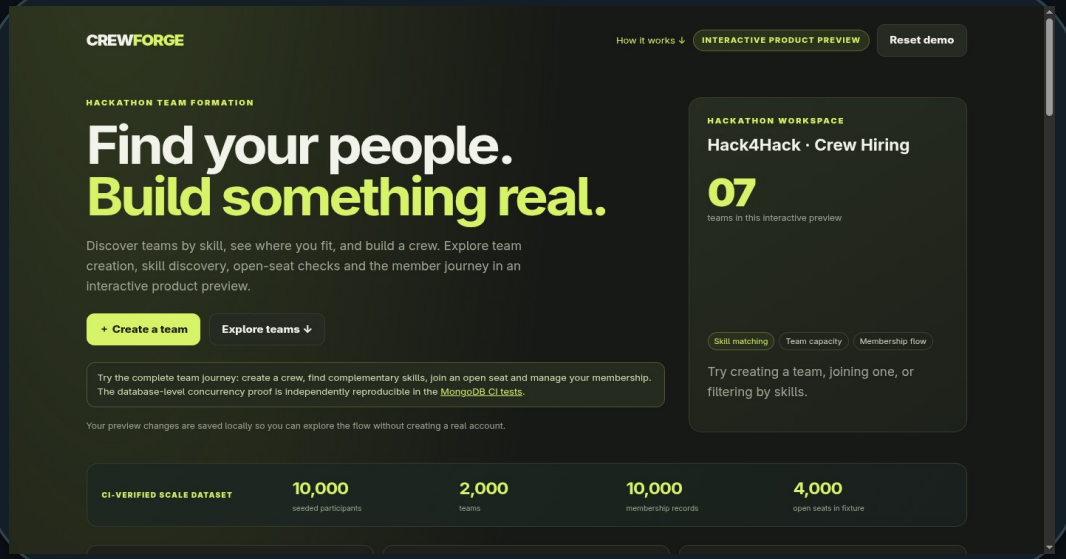
MongoDB change streams → bounded Server-Sent Events · leader succession · invite revocation · last-member team closure

Verified in code and automated tests: server-side role checks · idempotent requests · hashed invites · transactional audit events · reconnectable live updates.

08 / EVIDENCE

What judges can inspect and verify

A working product journey backed by transaction-level correctness and reproducible CI.



WORKING PRODUCT FLOWS

01 Create teams and define skill needs

02 Discover teams by skills and open seats

03 Submit idempotent join requests

04 Leader approval reserves the final seat atomically

05 Manage members, invites and team lifecycle

10,000
seeded participants

2,000
teams

10,000
membership records

50 → 1
concurrent final-seat race

CI: 65/65 tests · 97.43 req/s · 20.92 ms p95 · 0% HTTP failures · 6,000/6,000 board checks. Synthetic CI runner; not production capacity.

Repository: github.com/PSP-YN/Race-proof-team-formation · CI 41873925487 · CodeQL 41873925485